SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	***Shaik Abdul Avez***	**Reg. No.:**	***192472260***
Section / Batch:	**B.E CSE(AI)**	**Date:**	**13-10-2026**

Code of Conduct

*I **Shaik Abdul Avez(192472260)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

13-10-2026 **CO -1 AT-2** **Name : Shaik Abdul Avez**
ITA0329 **Reg No :192472260**

Q1. Full-Duplex Voice Channels from Spectrum Allocation

• Formula: Bandwidth = 45 MHz = 45,000 kHz. Full-duplex needs 2 simplex

channels of 25 kHz each = 50 kHz per channel.

• Calculation: 45,000 kHz / 50 kHz

• Answer: 900 channels

Q2. Single-Cell Channel Allocation in a Reuse Cluster

• Formula: Channels per cell = Total Channels / Cluster Size (N)

• Calculation: 560 / 7

• Answer: 80 channels

Q3. Simultaneous Voice Channels in GSM Deployment

• Formula: Total Channels = Number of Carriers * Time Slots per Carrier

• Calculation: 100 carriers * 8 time slots

• Answer: 800 voice channels

Q4. Decibel-Based Power Control Ratio

• Formula: Ratio in dB = 10 * log10(P1 / P2)

• Calculation: 10 * log10(80 / 4) = 10 * log10(20)

• Answer: 13.01 dB

Q5. Carrier-to-Interference Ratio (C/I)

• Formula: C/I (dB) = Signal (dBm) - Interference (dBm)

• Calculation: (-60 dBm) - (-92 dBm) = -60 + 92

• Answer: 32 dB

Q6. Channels per Cell in 42 MHz Spectrum

• Formula : Total channels = 42 MHz / 300 kHz. Then divide by cluster size 7.

• Calculation: (42,000 kHz / 300 kHz) / 7 = 140 / 7

• **Answer: 20 channels per cell**

Q7. Received Power with Path-Loss Exponent

• Formula: P proportional to d^(-n) where n = 4 and distance doubles

(factor of 2). Attenuation = 2^4 = 16.

• Calculation: 64 W / 16

• Answer: 4 W

Q8. Transmit Power Scaling for Doubled Cell Radius

• Formula: Power scales with R^n for inverse-square law (n = 2) Scaling factor = (2)^2.

• Calculation: 2^2

• Answer: Factor of 4

Q9. Total Channel Count Across Cluster

• Formula: Cluster Channels = Cluster Size * Channels per Cell

• Calculation: 4 * 15

• Answer: 60 channels

Q10. Doppler Shift Estimation

•Formula: fd = (v / c) * f, where v = 100 km/h = 27.78 m/s, f = 900

MHz, c = 3 * 10^8 m/s.

• Calculation: (27.78 / 3*10^8) * (9*10^8)

• Answer: 83.34 Hz

Q11. TDMA Time Slot Duration

• Formula: Slot Duration = Frame Duration / Number of Slots

• Calculation: 5 ms / 8

• Answer: 0.625 ms (625 microseconds)

Q12. Carrier Frequencies from Bandwidth Allocation

• Formula: Total Bandwidth / Carrier Spacing

• Calculation: 24,000 kHz / 200 kHz

• Answer: 120 carrier frequencies

Q13. Received Power in dBm (Path Loss)

• Formula : Transmit power = 8 W = 39.03 dBm. Subtract path loss.

• Calculation: 39.03 dBm - 33 dB

• Answer: 6.03 dBm

Q14. Free-Space Path Loss (FSPL)

• Formula: FSPL (dB) = 20 log10(d) + 20 log10(f) + 32.44 (d in km, f in MHz)

• Calculation: 20 log10(2.5) + 20 log10(1900) + 32.44 = 7.96 + 65.57 + 32.44

• Answer: 105.97 dB

Q15. Cluster Call-Handling Capacity

• Formula: Total Capacity = Number of Cells * Capacity per Cell

• Calculation: 6 * 250

• Answer: 1,500 simultaneous calls

Q16. Active Calls via Little's Law

• Formula: L = lambda * W, where lambda = 4000 calls/s, W = 180 s (3mins).

• Calculation: 4000 * 180

• Answer: 720,000 active calls

Q17. Base Station Total Bandwidth Consumption

• Formula: Total Bandwidth = Channels * Bandwidth per Channel

• Calculation: 150 * 25 kHz

• Answer: 3,750 kHz (3.75 MHz)

Q18. Guard Band Waste Calculation

• Formula : For 30 adjacent bands, there are 29 guard band interfaces.

• Calculation: 29 * 40 kHz

• Answer: 1,160 kHz (1.16 MHz)

Q19. Handoff Frequency Estimation

• Formula: Crossing distance = diameter = 2 * 1.5 km = 3.0 km. Total distance in 1 hour = 90km.

• Calculation: 90 km / 3.0 km

• Answer: 30 handoffs

Q20. Network Hierarchy User Capacity

• Formula: Centers * Base Stations * Users per Base Station

• Calculation: 8 * 40 * 250

• Answer: 80,000 users

Q21. Cell Count for Regional Coverage

• Formula: Hexagonal cell area factor approx 2.6 * R^2 = 2.6 * (0.4 km)^2 = 0.416 sq km.

• Calculation: 400 sq km / 0.416 sq km

• Answer: 962 cells

Q22. FDMA Channel Count from Total Spectrum

• Formula : Total Spectrum / Channel Bandwidth

• Calculation: 6,000 kHz / 30 kHz

• Answer: 200 channels

Q23. Offered Traffic in Erlangs

• Formula: A = lambda * h, where lambda = 1200 * 4 = 4800 calls/hr, h = 2.5/60 hrs.

• Calculation: 4,800 * (2.5 / 60)

• Answer: 200 Erlangs

Q24. Maximum Concurrent Users from Bandwidth

• Formula: Total Bandwidth / Bandwidth per User

• Calculation: 40,000 kHz / 400 kHz

• Answer: 100 users

Q25. Total Network User Capacity

• Formula: Stations * Users per Station

• Calculation: 1,200 * 600

• Answer: 720,000 users